

Detecting peptides outside the database using *de novo* sequencing and database search

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Most proteomics mass spectrometry analysis pipelines take as input a collection of observed mass spectra and aim to produce as output a corresponding list of peptides or proteins, along with corresponding quantifications. In this paradigm, the implicit hypothesis being tested is that “Peptide x was responsible for generating the observed spectrum y .” Downstream analysis then focuses on questions such as whether the observed peptide or protein quantifications differ among experimental groups. Typically, the core of such a pipeline involves searching the observed spectra against a provided protein database, although alternative approaches that involve *de novo* sequencing or some combination of *de novo* and database search are also possible.

In a metaproteomics setting, the presence or absence of a particular peptide in the sample is not always of primary interest. When studying, for example, the microbial population in a set of samples from ocean water, soil, or the human gut, two general classes of hypotheses are common: hypotheses regarding protein function **Uri: I am not sure I understand this correctly, I bring it up mostly in relation to the validity of our argument at the end of this paragraph: if by dynamics you also need quantification then you also need to know the actual spectrum, right? Re-worded.** —**Bill** and questions about the distribution of taxonomic groups in the microbial population. In either of these cases, we argue, the primary hypothesis should not be framed as “Peptide x was responsible for generating the observed spectrum y .” Instead, the question is whether “The peptide that was responsible for generating the observed spectrum y is identical or homologous to peptide x .”

To understand the motivation for this shift in hypotheses, consider the setting in which a metaproteomics sample is subjected to *de novo* sequencing analysis. In practice, the list of peptides that are produced by this analysis may be used to test hypotheses about the associated functional annotations, as represented, for example by KEGG or GO annotations. Alternatively, that same list may be subjected to taxonomic analysis by mapping the peptides onto taxonomic groups. In either of these settings, a truly novel peptide that has never been seen before is not of much use. Such a peptide cannot be associated with a protein ID and hence cannot be mapped to a functional label or a taxonomic group. On the other hand, if that novel peptide p can be mapped onto a known peptide p_n using a sequence similarity tool such as BLAST [1], then it may still be possible to use functional or taxonomic annotations associated with p_n [8, 7, 9].

Insert paragraph about related work.

In this work, we propose a two-pronged analysis strategy that subjects a given collection of metaproteomics data to both a database search and *de novo* sequencing, with the aim of producing a list of peptides that are identical or homologous to peptides in the sample (Figure 1). The procedure involves merging the peptides detected by the two methods into a single list and then searching that list against a protein database using a sequence alignment algorithm. Finally, the peptides resulting from this similarity search are subjected to a rigorous false discovery rate (FDR) control procedure, called RESET [4], which makes use of scores from the database search, the *de novo* sequencing, and the sequence alignment. Critical to the success of this approach is the inclusion of randomized decoys in the protein database, which enable RESET to accurately estimate the FDR in the combined list.

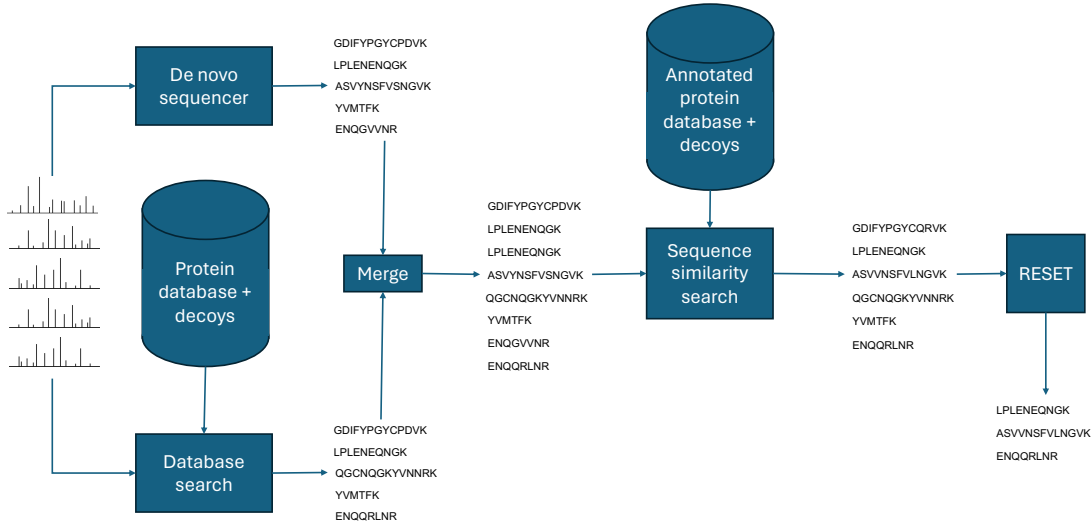


Figure 1: **Two-pronged analysis strategy.**

1 Methods

1.1 The denovo-db protocol

The denovo-db¹ procedure involves four steps (Figure 1), which we describe in detail here.

1.1.1 Step 1: Spectrum database search

The database search step is carried out with Comet [3], using a database containing peptides generated from metagenomic sequencing of the same samples (“metagenomic database”). Rather than using Comet’s built-in functionality to generate decoys, which involves independently reversing each target peptide, we generate decoy proteins in advance by reversing each protein sequence. This is done to guarantee that we can synchronize between the decoys used in this step and those used in Step 3 below. A concatenated target+decoy fasta file is then provided as input to Comet. We run Comet using semi-tryptic digestion without proline suppression of cleavage, allowing for up to three missed cleavages per peptide. We include one static modification, carbamidomethylation of cysteine (C+57.021464), and one variable modification, oxidation of methionine (M+15.9949, maximum three per peptide). The precursor m/z window is set to 10 ppm, and we allow up to three isotope errors. **Uri: We allow isotope errors up to +3 isotopic peaks?** The fragment m/z axis is discretized with bins of 0.02 m/z . All other Comet parameters are set to their default settings. We retain a single, top-scoring target or decoy peptide-spectrum match (PSM) for each spectrum by setting num_output_lines to 1. We then reduce this set of PSMs to a list of distinct peptide sequences, with modifications removed, retaining only the maximum score per peptide.

Uri: Note that if we want to save time we can only include target and decoy Comet PSMs that are above some FDR cutoff.

1.1.2 Step 2: *De novo* sequencing

For *de novo* sequencing, we use Casanovo version 5.0 [11]. We run the Casanovo “sequence” command using default settings. Casanovo uses a fixed set of seven types of variable modifications (methionine oxidation, asparagine deamidation, glutamine deamidation, N-terminal acetylation, N-terminal carbamylation, N-terminal NH₃ loss, and the combination of N-terminal carbamylation and NH₃ loss). We effectively disable Casanovo’s precursor m/z filtering by setting the precursor mass tolerance to a very large value. Casanovo

¹Placeholder name

Category	Feature	Type
DIAMOND	Database peptide length	integer
DIAMOND	Maximum sequence similarity score	real
DIAMOND	Maximum percent identity	real
DIAMOND	Number of distinct Casanovo peptides mapped to this peptide	integer
DIAMOND	Number of distinct Comet peptides mapped to this peptide	integer
Casanovo	Number of matched spectra	integer
Casanovo	Maximum score	real
Casanovo	Delta m/z	real
Casanovo	Number of distinct peptidoforms	integer
Comet	Number of matched spectra	integer
Comet	Tryptic cleavage at N-terminus	Boolean
Comet	Tryptic cleavage at C-terminus	Boolean
Comet	Maximum score	real
Comet	Delta m/z	real
Comet	Number of distinct peptidoforms	integer

Table 1: **Features input to RESET.** Each entry corresponds to a database peptide and its associated query peptide. The query peptide may have been detected in the database search, *de novo* sequencing, or both. The delta m/z values are in units of ppm, and are adjusted to account for isotope errors.

reports a single, top-scoring PSM for each spectrum. We then collapse this list to the maximum score per distinct, unmodified peptide.

Should we use some kind of Casanovo threshold? —Bill

1.1.3 Step 3: Sequence homology searching

The lists of peptides nominally detected by Comet and Casanovo are merged into a single list, and this list is searched against a second database using DIAMOND [2]. This second database is annotated, meaning that each database protein has an associated functional or taxonomic label. As before, we create a target+decoy database via protein reversal. Consequently, peptides that are shared between this database and the one used by Comet also share their decoys [Mike] Add details about how the search is run. For each query, we retain the top-scoring database peptide (or peptides, in the case of ties), along with the associated protein identifiers.

1.1.4 Step 4: RESET

In the final step of denovo-db, the database peptides output by DIAMOND are provided as input to the RESET algorithm [4], each coupled with a feature vector that is used by RESET to carry out FDR control. Each of the database peptides output by DIAMOND is matched to a query peptide that was produced by the Comet search, by Casanovo, or by both. Accordingly, the feature vector contains three different categories of features: those that are properties of the DIAMOND match, the Comet peptide, and the Casanovo peptide (Table 1). Database peptides that do not map to either a Casanovo peptide or a Comet peptide are discarded. RESET uses the given feature vectors to train a machine learning model to rank the peptides and use its estimating decoys to set the threshold for reporting discoveries at the desired FDR level ([4]).

Although many of the features listed in Table 1 are self-explanatory, some of them require additional explication. For example, it may be the case that, in the sequence homology search, multiple Comet or Casanovo peptides match to a single database peptide. For matches to Casanovo predictions, we record the highest Casanovo score as well as the number of distinct Casanovo peptides. The case of multiple Comet peptides matching to a single peptide in the homology search is trickier because the Comet peptides can be a mixture Comet target and decoy peptides. Still, we chose to allow those in light of transitivity: if Comet found that the spectrum well-matches a decoy peptide and that decoy peptide is in turn similar to a target peptide in the larger DB, then it is not unreasonable to attribute the match to the latter. Hence, we count the total number of Comet matches regardless of whether they are targets or decoys.

Some additional notes about these features are as follows:

- For Comet, the number of matched spectra is computed considering only the top-scoring PSM per spectrum.
- For both Casanovo and Comet, the delta m/z feature is calculated while taking into account potential isotope errors. In particular, offsets up to three C_{13} masses are considered, and the one that yields the smallest ppm error is reported.
- The counts of “peptidoforms” refer to variants of a peptide sequence annotated with charge state and localized PTMs. For Comet, the count comes from the set of top-scoring PSMs per spectrum.

1.2 Competing methods

We compare denovo-db to two alternative methods for producing a list of annotated peptides from a metaproteomics dataset.

The first method directly maps the observed spectra to peptides in the annotated protein database. For this step, we use the same Comet database search procedure, applied to the annotated rather than the metagenomic database. This search is followed by post-processing with Percolator [5] version **X.XX** using default parameters, producing a list of peptides subject to Percolator’s 1% peptide-level FDR control. Modifications and duplicates are removed from this list. **Uri: maybe it’d make more sense to use Percolator-RESET to even the playing field? I was assuming that we are using the latest version of Percolator, which indeed implements Percolator-RESET. Mike, can you verify this? —Bill**

The second method is modeled after the approach implemented in MetaGOMics. First, we use Comet to search the metaproteomics data against the metagenomics database, as in the first step of denovo-db. This search is followed by Percolator post-processing, thereby producing a list of peptides subject to 1% FDR control, as above. The resulting list is split into two groups: peptides that appear in the annotated database and those that do not. The latter set of peptides is searched against the annotated database using NCBI BLAST [1], with an E-value cutoff of 10^{-10} and retaining only the top hit. The peptides output by BLAST are then added to the list of annotated peptides from the database search step.

1.3 Validation

We validate our FDR control procedure in two ways. First, we augment the annotated database with an additional 10% of “entrapment” protein sequences. These entrapment proteins are created by selecting a random (target or decoy) protein from the annotated database, identifying tryptic sites, and then independently shuffling each peptide between tryptic sites. In keeping with the overall approach, we also inject 15% of these entrapment proteins into the metagenomic database. **Uri: Alternatively, we can use a similar fraction to what we find in the real data based on Comet searches.** As in our recent work [10], we can use the rate at which these entrapment sequences are selected to estimate the false discovery proportion (FDP) and then compare the FDP to the FDR threshold provided to RESET. Specifically, if at a given score threshold we accept n peptides, of which n_e are entrapments, then the estimated FDP is $(10+1)n_e/n$, where the factor of 10 corresponds to the 10%.

Second, we empirically test the equal chance assumption that underlies all target-decoy analyses [6], which states that an incorrectly matched spectrum is equally likely to match a target or a decoy. We do this by creating a synthetic spectrum from a decoy peptide, using AlphaPeptDeep [12] to select fragment ion intensities and adding simulated noise. **Ideally, get simulated noise from “blank” spectra. —Bill** Running such spectra through our procedure, we can empirically verify that they have a 50/50 chance of hitting a target or the decoy protein in the annotated database.

- **Use a mouse sample.**
- **Run our procedure using (1) 75% of the mouse proteome as the first database and (2) swissprot minus any mouse-specific peptides as the second database.**
- **Find the most specific taxonomic ID for each protein in which each peptide appears.**

- Add a $+1/n$ to each of those IDs, where n is the number of proteins in which this peptide appears.
- For each taxonomic ID, calculate its taxonomic distance to mouse (i.e., number of edges along the taxonomy tree).

2 Combatting memorization

One concern associated with any machine learning model is the possibility that the model will overfit to the training data, effectively memorizing features of that specific data set. When this happens, the model’s performance on unseen test data deteriorates. More tellingly, in the context of FDR control, overfitting can potentially lead to a bias in favor of target peptides, which are more likely to resemble the training data.

To combat this problem, we construct training data for Casanovo that prevents the model from distinguishing between targets and decoys. This is done by training the model with a 50/50 mixture of target PSMs and reversed decoy PSMs. To create this training set, we randomly split the PSMs in the training data at the peptide level into target and decoy subsets. PSMs in the target subset remain unchanged. Each PSM in the decoy subset involving a peptide p and a spectrum s is replaced by a “reversed PSM,” generated as follows:

1. Reverse the peptide p to generate p_{rev} , leaving the C-terminal amino acid in place.
2. Use a pre-trained model, [AlphaPeptDeep?](#), to predict the fragment ion intensities associated with peptide p and the reversed peptide p_{rev} . This step includes b- and y-ions as well as neutral losses of water and ammonia [plus maybe others?](#) and yields two theoretical spectra, $s(p)$ and $s(p_{rev})$.
3. Match the peaks in the observed spectrum s to those of $s(p)$ using a specified fragment ion tolerance (we use 0.05 m/z).
4. Use this matching to segregate the peaks in the observed spectrum s and the predicted spectrum $s(p)$ into two categories: “matched” and “unmatched.” Unmatched peaks in the observed spectrum s presumably correspond to noise, whereas unmatched peaks in the predicted spectrum $s(p)$ correspond to missing fragmentation events.
5. Separately sort the peaks in the two predicted spectra, $s(p)$ and $s(p_{rev})$, by decreasing intensity.
6. Traverse both peak lists simultaneously. Each time we encounter an unmatched peak in $s(p)$, we remove the corresponding peak from $s(p_{rev})$. Each time we encounter a matched peak in $s(p)$, we compute the delta mass between the observed peak in s and the theoretical peak in $s(p)$. This delta mass is, by definition, in the range $[-0.05, 0.05]$ m/z . We then perturb the corresponding peak in $s(p_{rev})$ by this delta mass. After this traversal, the number of matched peaks in $s(p)$ will equal the number of peaks remaining in $s(p_{rev})$.
7. Add all the unmatched (noise) peaks in s to the above filtered version of $s(p_{rev})$.

The reversed PSM, consisting of the peptide p_{rev} and the processed spectrum $s(p_{rev})$, is then added to the training set.

3 Results

4 Discussion

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